

Assignment 5_Claxton_Funds of ML

2026-06-13

```
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.5.3
```

```
## Loading required package: ggplot2
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
## Loading required package: lattice
```

```
library(stats)
library(cluster)
```

```
## Warning: package 'cluster' was built under R version 4.5.3
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.3
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
Cereals <- read.csv("C:\\Users\\User\\OneDrive\\Desktop\\Grad School_Business Analytics\\Fundamentals of Data Science\\Cereals.csv")
colSums(is.na(Cereals))
```

```
##      name      mfr      type calories protein      fat  sodium  fiber
##       0        0        0         0        0         0         0         0
##    carbo  sugars  potass vitamins  shelf  weight     cups  rating
##       1        1         2         0        0         0         0         0
```

```
#look for missing values
```

```
CerealNum <- Cereals %>%  
  select(-name, -mfr, -type) %>%  
  na.omit()  
#remove rows with missing  
CerealNum <- Cereals %>%  
  select(-name, -mfr, -type)  
#only keeping numeric  
CerealNorm <- scale(CerealNum)  
#normalize
```

```
ag.single <- agnes(CerealNorm,  
  metric = "euclidean", method = "single")  
#single  
ag.complete <- agnes(CerealNorm,  
  metric = "euclidean",  
  method = "complete")  
#complete  
ag.average <- agnes(CerealNorm,  
  metric = "euclidean",  
  method = "average")  
#average  
ag.ward <- agnes(CerealNorm,  
  metric = "euclidean",  
  method = "ward")  
#ward  
ag.single$ac
```

```
## [1] 0.6023689
```

```
ag.complete$ac
```

```
## [1] 0.8387548
```

```
ag.average$ac
```

```
## [1] 0.7768921
```

```
ag.ward$ac
```

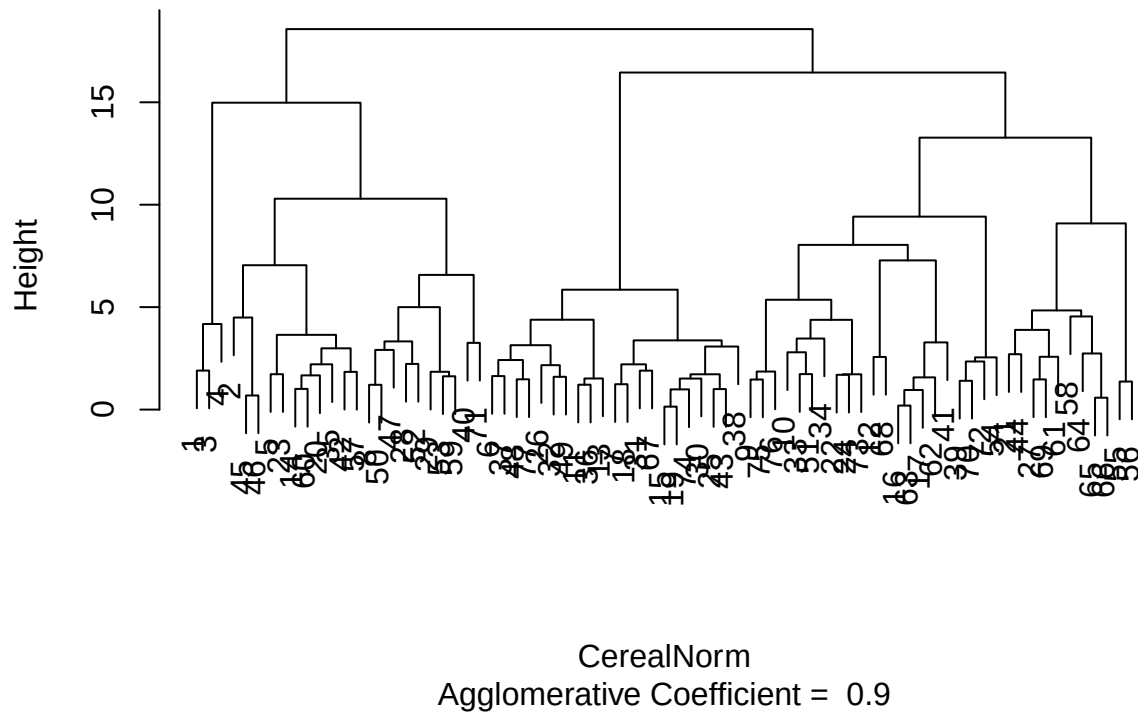
```
## [1] 0.9029485
```

```
#print agglomerative coefficients
```

Question #1: After running hierarchal clustering using single, complete, average, and ward linkages, ward has the strongest agglomerative coefficient measure

```
plot(ag.ward, which.plots = 2)
```

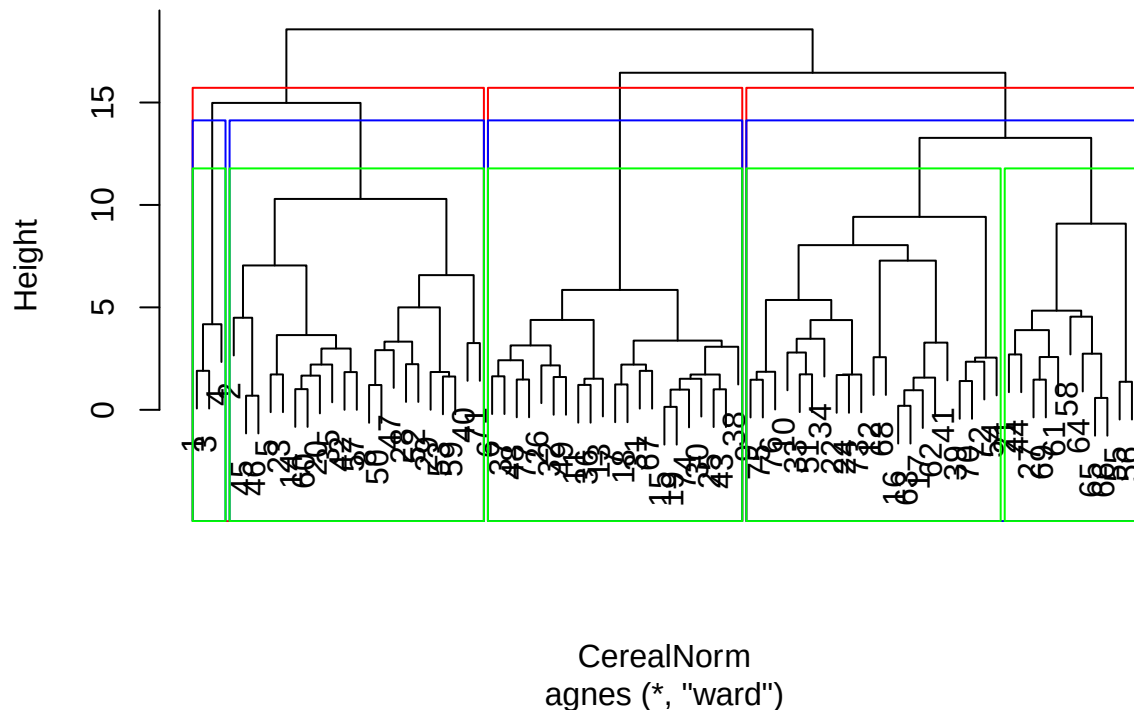
Dendrogram of `agnes(x = CerealNorm, metric = "euclidean", method = "ward")`



```
#plot dendrogram  
#this will tell us how many clusters  
#to choose. We are looking at the  
#height to determine where the big jump is  
#because height is = differences between clusters
```

```
pltree(ag.ward)  
  
rect.hclust(as.hclust(ag.ward), k = 3, border = "red")  
rect.hclust(as.hclust(ag.ward), k = 4, border = "blue")  
rect.hclust(as.hclust(ag.ward), k = 5, border = "green")
```

endrogram of `agnes(x = CerealNorm, metric = "euclidean", method = "ward")`



*#Note to self: when we choose a height, we are drawing
#a horizontal line. The number of vertical lines
#that horizontal line crosses is the number of
#categories*

```
hc <- as.hclust(ag.ward)

for(k in 2:8){
  cat("\nK =", k, "\n")
  print(table(cutree(hc, k)))
}
```

```
##
## K = 2
##
## 1 2
## 24 53
##
## K = 3
##
## 1 2 3
## 24 21 32
##
## K = 4
##
```

```
## 1 2 3 4
## 3 21 21 32
##
## K = 5
##
## 1 2 3 4 5
## 3 21 21 21 11
##
## K = 6
##
## 1 2 3 4 5 6
## 3 11 21 10 21 11
##
## K = 7
##
## 1 2 3 4 5 6 7
## 3 11 21 10 17 11 4
##
## K = 8
##
## 1 2 3 4 5 6 7 8
## 3 11 21 10 17 9 4 2
```

*#show the population of clusters for
#various k values*

Question 2: I am selecting to use 5 clusters. Based off the dendrogram, this is where the greatest differences between clusters starts to emerge (based off the changes in height). This K value will avoid both overfitting and over-generalization. It also has reasonably balanced cluster sizes.

```
set.seed(123)

train.index <- createDataPartition(
  1:nrow(CerealNorm),
  p = 0.5,
  list = FALSE
)

train.cereal.norm <- CerealNorm[train.index, ]
test.cereal.norm <- CerealNorm[-train.index, ]
#create data partition
```

```
ag.train <- agnes(
  train.cereal.norm,
  metric = "euclidean",
  method = "ward")

hc.train <- as.hclust(ag.train)

cluster.train <- cutree(hc.train, k = 5)
#clustering our training data
```

```
train.df <- as.data.frame(train.cereal.norm)
```

```
centroids <- aggregate(
  train.df,
  by = list(cluster = cluster.train),
  mean
)
#measure the centroids from training data
centroids
```

```
##   cluster  calories  protein    fat    sodium    fiber    carbo
## 1      1  1.1294209  0.4151897  0.7598724  0.1430396  0.49567976  0.1358223
## 2      2 -2.4062220  1.3286071 -0.5096880  0.4810160  3.92220915 -1.8689591
## 3      3 -0.1821886  0.4151897 -0.3440932  0.3352223  0.02948529  0.1642589
## 4      4  0.2784101 -0.8495421  0.2163817  0.1782137 -0.67697865 -0.6582155
## 5      5 -1.0131461  0.4151897 -0.5806572 -1.8791382  0.09808247      NA
##      sugars    potass  vitamins    shelf    weight    cups
## 1  0.6283804  0.87897074  0.8492984  0.9515734  1.3095459  0.01940817
## 2 -1.0337225  3.21447648 -0.1453172  0.9515734 -0.1967771 -1.74478214
## 3 -0.4373955      NA -0.1453172  0.4177217 -0.1967771 -0.28138747
## 4  1.0304864 -0.70757084 -0.1453172 -0.5267853 -0.1967771  0.16411328
## 5      NA -0.02164175 -0.7847130 -0.5927834 -0.8328505  0.24108296
##      rating
## 1 -0.4073416
## 2  2.4132417
## 3  0.3090034
## 4 -1.0785670
## 5  1.3265700
```

```
centroid.matrix <- as.matrix(centroids[, -1])
test.matrix <- as.matrix(test.cereal.norm)
#Get the euclidean distance from each
#point in test subset to the centroids from
#train set
ag.full <- agnes(
  CerealNorm,
  method = "ward")

hc.full <- as.hclust(ag.full)

cluster.full <- cutree(hc.full, k = 5)
```

```
distance.matrix <- as.matrix(
  dist(rbind(centroid.matrix, test.matrix)))
#distance matrix containing centroid
#to centroid, test to centroid,
#and test to test
```

```
test.centroid.distances <-
  distance.matrix[
    (nrow(centroid.matrix)+1):nrow(distance.matrix),
    1:nrow(centroid.matrix)]
#measure distances from test to centroid
```

```
test.assign <- apply(
  test.centroid.distances,
  1, which.min)
#assign test data to each centroid
```

```
full.test <- cluster.full[-train.index]
table(test.assign, full.test)
```

```
##           full.test
## test.assign  1  2  3  4  5
##           1  0  6  0  1  0
##           2  1  0  0  0  0
##           3  0  4  0 13  0
##           4  0  0  8  0  0
##           5  0  0  0  0  4
```

```
mean(test.assign == full.test)
```

```
## [1] 0.1081081
```

Question #3: The clusters of the test set had only 10.8% agreement with the HC model on the full data set. This is much lower than I expected and indicates an error in the calculation. After researching, I suspect there may be a labeling issue but I'm not quite sure how to resolve it.

Question #4: Normalization would not be good if we are trying to cluster cereals by health benefits. Calories, sugar, protein, and fiber are all important attributes and should retain their original scale for cluster analysis. This way, cereals higher in fat, sugar, calories, etc will be grouped together and those higher in fiber and protein will fall into a different (healthy) cluster. Of course the key to this is parameter selection and model evaluation.